

Claim-Gated Evaluation for Small Retrieval Deltas: A Confirmatory Mixed-Domain Study

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Abstract

This paper presents an evaluation methodology for small retrieval improvements where overclaim risk is high. We run a confirmatory paired test of Kalman-style uncertainty fusion versus mean fusion on a mixed-domain benchmark and focus on decision protocol rather than model novelty. Under the tested configuration ($n_{\text{pairs}} = 1193$, `max_candidates=100`), Kalman-minus-mean Δ nDCG@10 is -9.2588×10^{-6} , Holm-adjusted $p = 1.0$, and recall@100 delta is 0.0. Latency is higher (ratio 1.0723) at matched FLOPs (ratio 1.0). We therefore report a constrained negative result: no reliable benefit is demonstrated under this protocol. The contribution is a reusable claim-gated template with paired inference, baseline-matrix reporting, and artifact-backed reproducibility.

1 Introduction

Modern IR papers often report tiny metric deltas that are sensitive to sampling and benchmark composition. This submission targets that evaluation problem directly: how should one decide whether a small observed improvement is claim-worthy?

Our confirmatory question is narrow: does Kalman-style uncertainty fusion improve nDCG@10 over mean fusion in a domain-balanced mixed-domain setting? We use a predeclared claim gate: positive direction, practical effect size, multiplicity-adjusted significance, and acceptable efficiency overhead must all pass.

Scope of claim. We do **not** claim Kalman fusion is generally ineffective. We claim only that this study did not demonstrate a reliable Kalman-over-mean benefit under the tested protocol.

2 Contributions

3 Protocol and Statistical Framework

Primary endpoint: paired query-level Δ nDCG@10 (Kalman minus mean). Secondary endpoints are Δ recall@100, latency ratio, and FLOPs ratio.

Design: domain-balanced paired evaluation so each query is scored by both methods; confirmatory and exploratory slices are separated.

Inference: paired Wilcoxon testing, Holm correction, and bootstrap interval summaries.

Decision rule: a positive claim requires (i) Δ nDCG@10 > 0 , (ii) $|\Delta| \geq 0.02$, (iii) Holm-adjusted $p < 0.05$, and (iv) efficiency constraints passing.

Table 1: Contributions of this TMLR submission.

Contribution	What is delivered
Evaluation protocol	Claim-gated exploratory/confirmatory split with domain-balanced paired queries and predeclared acceptance rules.
Paired statistical framework	Query-level paired deltas, bootstrap uncertainty summaries, and Holm correction for multiplicity-aware decisions.
Baseline matrix	Canonical multi-baseline comparison including mean fusion, weighted mean, hard routing, learned combiner, and single generalist.
Negative result	No reliable Kalman-over-mean gain under confirmatory criteria, with explicit gate-blocked verdict.
Reusable artifact template	Registry-backed manuscript fields and table includes to reduce manual number transcription.

Table 2: Baseline matrix (canonical_benchmark_v3_fast_1200 artifact family). Metrics are means from the canonical summary export.

Method	nDCG@10	latency (ms)
KalmanorixFuser	0.3666	57.319
MeanFuser	0.3666	53.457
Fixed weighted mean	0.3666	53.753
Hard routing (top-1)	0.3489	35.972
Learned linear combiner	0.3666	53.895
Single generalist	0.3406	27.595

4 Results

Method	nDCG@10	MRR@10	Recall@10	latency (ms)	FLOPs proxy
KalmanorixFuser	0.0941	0.1025	0.1358	202.858	3.000
MeanFuser	0.0943	0.1022	0.1362	198.558	3.000
fixed weighted mean baseline	0.0940	0.1018	0.1363	2.004	3.000
hard routing baseline	0.0979	0.1147	0.1299	6.433	1.000
learned linear combiner	0.0934	0.1058	0.1334	1.942	2.000
single generalist model	0.0952	0.1090	0.1287	2.108	1.000
all-routing + mean baseline	0.0940	0.1018	0.1363	2.043	3.000

The confirmatory result is gate-blocked: Δ nDCG@10 is near zero (-9.2588×10^{-6}), adjusted significance is absent ($p = 1.0$), and recall change is zero. The run remains sufficiently powered for the tested setup ($n_{\text{pairs}} = 1193$), so the correct interpretation is *no demonstrated reliable benefit*, not “insufficient data.”

5 Why Did Kalman Not Help Here?

The artifacts support five bounded explanations:

1. **Uncertainty signal weakness.** Query-level uncertainty features did not produce stable ranking gains over a strong mean baseline.

2. **Limited downstream effect of calibration.** Calibration analyses showed little translation from uncertainty adjustment to retrieval metric movement.
3. **Covariance richness added latency.** Richer covariance handling increased runtime overhead without measurable quality gain.
4. **Mean fusion remained competitive.** Baseline-matrix results show near-tied quality against simpler fusion.
5. **Candidate cap and benchmark scope.** With `max_candidates=100` and benchmark `mixed_beir_v1.2.0`, conclusions are protocol-local and may differ at other budgets or domains.

6 Reviewer-Facing Limitations

- **Narrow claim target.** The evidence supports only a within-protocol statement, not a universal statement about Kalman methods.
- **Configuration boundedness.** Results are tied to this implementation, this candidate cap, and this mixed-domain benchmark family.
- **External validity still open.** Different retrievers, uncertainty estimators, or larger candidate pools may change outcomes.
- **Single confirmatory track.** Additional independent confirmatory reruns would strengthen robustness.

7 Reproducibility Statement (TMLR)

All numerical claims in this manuscript are sourced from committed artifacts, primarily `results/evidence_registry.json` and canonical benchmark summaries. Manuscript constants are included via `paper/tmlr/includes/generated_metrics` to avoid manual transcription. The baseline matrix is exported from canonical benchmark artifacts (v3 fast-local family).

To reproduce the paper build: `cd paper/tmlr; pdflatex main.tex; bibtex main; pdflatex main.tex; pdflatex main.tex`. To reproduce repository checks: `ruff format --check .` and `pytest -m "not integration and not stress"`.

8 Conclusion

This TMLR submission is about methodology: a claim-gated evaluation protocol for small retrieval deltas. In the tested mixed-domain confirmatory setting, Kalman-style fusion did not show a reliable advantage over mean fusion. The recommended scientific conclusion is constrained and artifact-backed: no reliable benefit was demonstrated under the tested protocol.

References